

PARSONS TRI-CITY, KS, USA

WMO: 745431

Lat: 37.328N

Lon: 95.504W

Elev: 869

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.2	-0.7	5.5	12.4	3.6	6.8	15.9	25.2	41.3	22.6	42.0	5.6	0	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	98.9	75.2	95.3	76.1	92.4	75.6	79.5	91.0	78.2	89.9	77.2	88.7	8.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	141.7	86.1	74.9	135.3	84.8	73.5	128.6	83.2	43.6	91.1	42.4	90.3	41.2	88.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.8	19.1	17.0	-1.2	101.6	7.5	4.5	-6.6	104.9	-11.0	107.5	-15.2	110.0	-20.7	113.3		
			-1.0	81.6	8.2	1.4	-6.9	82.6	-11.7	83.4	-16.4	84.2	-22.3	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	34.2	38.2	48.1	57.9	67.1	76.0	80.4	79.4	71.7	59.4	47.2	36.7
	DBStd	18.53	10.85	11.25	10.80	9.03	7.65	5.56	5.30	5.74	7.46	9.22	9.95	10.47
	HDD50	1666	499	356	165	31	1	0	0	0	0	22	166	426
	HDD65	4215	954	750	530	245	70	3	0	1	24	224	537	877
	CDD50	4638	10	25	107	268	531	780	943	912	651	313	83	15
	CDD65	1708	0	0	7	32	135	333	478	447	224	49	3	0
	CDH74	18041	0	6	56	281	1155	3395	5481	4994	2221	429	23	0
	CDH80	8164	0	0	7	47	322	1403	2773	2573	934	104	1	0
Wind	WSAvg	6.8	7.6	7.8	8.5	8.6	6.8	6.4	5.1	4.8	5.1	6.3	7.3	7.3
Precipitation	PrecAvg	41.7	1.4	1.5	2.8	4.3	6.0	5.9	3.9	3.4	4.2	3.5	2.6	2.1
	PrecMax	58.5	4.3	4.6	6.7	11.8	10.6	18.1	9.1	7.8	9.7	9.3	7.3	5.6
	PrecMin	24.5	0.1	0.0	0.4	0.7	2.4	1.6	0.9	0.2	0.9	0.3	0.0	0.2
	PrecStd	7.6	0.9	1.2	1.8	2.4	2.5	3.5	2.0	1.7	2.1	2.2	2.0	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.6	73.5	79.8	85.0	90.0	96.6	103.0	103.5	98.5	87.5	77.3	68.8
		MCWB	54.1	57.8	63.5	67.4	71.9	77.0	74.3	75.4	72.7	71.1	62.1	60.7
	2%	DB	62.2	67.3	74.2	80.9	86.7	92.7	98.9	99.3	92.0	82.7	72.3	63.2
		MCWB	52.2	55.8	60.7	65.0	71.5	75.9	76.0	74.8	74.2	67.4	59.6	55.1
	5%	DB	57.0	62.7	70.2	77.2	83.6	90.3	95.2	95.8	89.5	79.0	68.1	57.5
		MCWB	47.5	53.0	57.6	63.2	70.6	75.6	76.8	75.5	73.4	64.9	58.3	50.4
	10%	DB	51.8	56.9	66.1	73.4	80.9	87.9	92.1	92.2	85.6	74.8	63.7	53.7
		MCWB	44.0	48.2	56.4	61.9	69.6	74.3	76.6	74.8	71.4	64.1	55.1	47.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	62.3	66.2	70.4	76.0	80.1	81.5	81.2	78.2	73.7	66.0	62.6
		MCDB	63.3	69.2	76.2	80.6	84.6	91.3	91.6	90.9	89.2	82.8	73.0	67.4
	2%	WB	53.6	58.7	63.2	68.0	74.3	78.4	79.8	78.9	76.4	71.3	62.7	57.2
		MCDB	60.2	65.4	70.9	76.8	83.1	89.4	91.0	91.0	87.5	78.4	68.8	61.7
	5%	WB	48.6	54.0	60.5	66.1	72.5	77.1	78.4	77.5	75.0	69.0	59.9	52.5
		MCDB	56.1	60.5	67.7	74.1	81.1	87.6	90.0	90.0	85.5	74.8	66.2	56.2
	10%	WB	44.3	48.3	57.3	63.7	70.8	75.5	77.5	76.2	73.2	66.1	56.7	47.3
		MCDB	51.2	56.7	64.6	71.7	78.8	85.1	89.2	88.2	82.6	72.7	62.8	52.5
Mean Daily Temperature Range	5% DB	MDBR	21.1	21.4	22.1	21.6	20.2	19.6	20.4	21.9	22.5	23.1	22.2	20.0
		MCDBR	29.6	29.0	27.4	25.3	22.4	21.7	24.2	26.4	25.0	26.2	26.6	26.7
		MCWBR	19.7	19.3	16.0	14.1	11.0	9.3	7.7	8.2	10.2	13.4	16.3	18.6
	5% WB	MCDBR	25.7	25.8	21.3	20.5	18.9	19.4	19.8	21.2	21.0	19.2	22.1	21.3
		MCWBR	18.6	19.1	14.5	13.6	10.7	9.5	8.4	9.0	9.4	11.9	15.8	17.9
Clear-Sky Solar Irradiance	taub	0.298	0.310	0.349	0.382	0.418	0.424	0.438	0.442	0.387	0.346	0.310	0.290	
	taud	2.472	2.452	2.382	2.318	2.275	2.283	2.290	2.255	2.374	2.467	2.509	2.513	
	Ebn at Noon	284	294	290	285	275	272	267	264	274	277	277	279	
	Edh at Noon	26	30	36	40	43	42	42	42	35	29	25	23	
All-Sky Solar Radiation	RadAvg	769	979	1284	1623	1812	2082	2076	1862	1556	1145	826	655	
	RadStd	73	154	78	109	168	180	137	119	120	138	92	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	-2.25	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page